

New Generation English B· Diagnostic Test

C2 (Proficiency Diagnostic) B· Reading

READING

Why simplicity in scientific theory is sometimes misleading

Few ideas in science enjoy as much rhetorical favour as simplicity. Researchers invoke Occam's razor; teachers cite Newton's elegant equations; popular writers admire those theories that can be condensed into a single sentence. There is, however, a quieter scientific tradition that holds simplicity to be a useful working preference rather than a reliable indicator of truth. According to this view, the universe has no obligation to be elegant, and the conviction that it must be can become its own intellectual hazard.

The argument is not that simplicity is unimportant. A theory with fewer arbitrary parameters is easier to test, easier to teach and harder to disguise as something it is not. These are genuine virtues. The problem appears when simplicity becomes a heuristic so deeply assumed that it operates as a hidden axiom when researchers reach instinctively for the simpler model because they believe nature must agree, rather than because the evidence has yet decided between competing alternatives.

Historical examples are instructive. Early models of planetary motion before Kepler were aesthetically attractive but explanatorily poor. Pre-genetic models of inheritance were similarly tidy. In both cases, the simpler description was, for a while, mistaken for the more correct description. Acceptance of the messier alternative required the field to tolerate temporary inelegance to live, intellectually, in a less polished house until the new theory could be properly furnished.

This lesson has methodological consequences. A research culture that punishes complexity as inelegant will systematically slow the discovery of phenomena whose actual structure happens to be complex. Biology is the obvious example. So is climate science, where the genuine answer involves dozens of variables behaving non-linearly, and where each attempt to reduce the picture to a clean slogan misrepresents the science it intends to summarise. Strangely, the strongest demand for simplification often comes from outside the discipline rather than from within it.

The wisest formulation, perhaps, is to treat simplicity as a default rather than a doctrine. Begin with simple models, prefer them when they fit, and abandon them without sentiment when the evidence does not cooperate. Refuse, however, to mistake parsimony for proof. A theory survives because the world appears to behave the way it predicts, not because it pleases our taste for tidiness. To remember this and to insist on it inside disciplines that are quietly drifting away from it is one of the small, ungrateful jobs that careful thinkers continue to do, often without applause.

1. What principle of simplicity does the writer mention by name?

- A) Pascal's wager.
- B) Occam's razor.**
- C) G del's theorem.

2. According to the "quieter scientific tradition", how should simplicity be regarded?

- A) As a useful working preference rather than a reliable indicator of truth.**
- B) As the central law of nature.

